INCH-POUND

MIL-C-44389(GL)
29 December 1989

MILITARY SPECIFICATION

CEREAL BARS, RATION LIGHTWEIGHT, 30-DAY

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cereal bars for use by the Department of Defense as a component of operational rations.

1.2 Classification. The products shall be of the following types and styles as specified (see 6.1).

Type I - Shredded wheat cereal bar
Type II - Corn flakes cereal bar
Type III - Wheat flakes cereal bar
Type IV - Malted wheat granules cereal bar
Type V - Bran flakes cereal bar
Type VI - Oat cereal biscuit bar

Style A - 1.0 by 3.0 by 0.6-inch bar weighing 32 to 38 grams
Style B - 2.0 by 3.0 by 0.6-inch bar weighing 64 to 76 grams

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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SPECIFICATIONS

FEDERAL

A-A-20000	-	Cereal, Prepared; Ready-to-Eat
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning, Material, Cellulosic
PPP-C-1752	-	Cushioning, Material, Packaging, Unicellular, Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

American Society for Testing and Materials (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue. N.W., Washington, DC 20418.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of

Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to the first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Cereals. Cereals except shredded wheat, shall be fortified in accordance with the Federal Food, Drug, and Cosmetic Act. At time of use the cereal shall contain no more than 3 percent moisture. The cereals shall comply with the following types, classes and styles of A-A-20000:

a. Shredded wheat biscuits, type I, class 1.
b. Wheat flakes, type I, class 3.
c. Bran flakes, type I, class 5.
d. Malted wheat cereal granules, type I, class 10 (see 6.4.1).
e. Corn flakes, type II, class 1.
f. Oat cereal biscuits, type IV, class 3, style A (see 6.4.2).

3.2.2 Cream substitute, dry, non-dairy. The cream substitute shall be a white, free flowing powder containing corn syrup solids, partially hydrogenated vegetable oils, sodium caseinate, mono and di-glycerides, dipotassium phosphate, artificial flavors and colors.

3.2.3 Flour, wheat. Wheat flour shall be free-flowing, enriched, general purpose white (bleached) flour.

3.2.4 Glycerine, food grade. Food grade glycerine shall comply with the requirements of the Food Chemical Codex.

3.2.5 Margarine. Margarine shall comply with the FDA Standard of Identity for Margarine (21 CFR 166.110). The margarine shall be manufactured from corn oil and shall possess a fine and pleasing flavor. The body shall be smooth, firm and homogeneous. The product shall possess a uniform medium yellow color. The margarine shall be salted.

3.2.6 Milk, nonfat dry (low heat). Nonfat dry milk shall be U.S. Extra Grade, Low Heat of the U.S. Standards for Grades of Nonfat Dry Milk (Spray Process).

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3.2.7 Salt. Salt shall be iodized, white, refined sodium chloride with or without anti-caking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.8 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof.

3.2.9 Titanium dioxide. Titanium dioxide shall be water dispersible and shall comply with the requirements of the Food Chemical Codex.

3.2.10 Water. Water used for sanitizing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Formulation and preparation of matrix. The matrix shall be formulated and prepared as follows:

Ingredients	Percent by weight
Cereal (see type specified)	37.50
Margarine	26.00
Flour, wheat	23.20
Sugar, white, granulated	13.00
Salt	0.30

a. Mix dry ingredients except cereal in a Hobart type mixer.

b. Add margarine and blend until margarine forms chunks about 1/4 inch in diameter.

c. Add one-half the cereal and mix well.

d. Blend in remaining cereal until just mixed. There should be intact pieces of cereal visible in the batter.

e. Pour matrix onto a cookie sheet to a depth of about 1/2 inch and bake at 250°F until moisture content is less than 2 percent. With proper thickness, baking should take 10 to 15 minutes. The baked matrix shall be light to medium tan in color with no areas that are brown, scorched, or burnt, except for corn flakes which may have a golden tan to tan color.

3.3.1.1 Handling of prepared matrix. Store matrix in vacuum sealed container until ready to use. Matrix may be held in accordance with good commercial practice for up to 30 days before forming into bars.

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3.4 Formulation, preparation, and compression of cereal bars. The cereal bars shall be formulated, prepared, and compressed as follows:

Ingredients	Percent by weight
Cereal matrix (see type specified)	86.0
Cream substitute, dry non-dairy	4.9
Milk, nonfat, dry	4.9
Glycerine	4.0
Titanium dioxide	0.2

a. Mix matrix, nondairy creamer, nonfat dry milk, and titanium dioxide in a suitable mixer until matrix is reduced to about 1/2 inch pieces. (Matrix may be ground through a screen not smaller than 1/2 inch prior to mixing).

b. Gradually, add glycerine and mix just until blended.

c. The cereal bar mixture shall be compressed so as to produce bars complying with the finished product requirements (see 3.6).

3.5 Pouch filling and sealing. The compressed bar(s) shall be packaged immediately in accordance with 5.1.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no compression streaks or stains.

e. The dimensions and weights shall be as follows:

Style A - Dimensions: 1.0 (± 0.1) inch by 3.0 (± 0.1) inches by 0.6 (± 0.1) inch thick. Weight: 32 to 38 grams.

Style B - Dimensions: 2.0 (± 0.1) inches by 3.0 (± 0.1) inches by 0.6 (± 0.1) inch thick. Weight: 64 to 76 grams.

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f. The moisture content shall be not greater than 5.0 percent.

g. No pouch shall contain a bar which has been broken or crumbled (more than three pieces).

h. The bar shall be easily crumbled into 50 mL of water and rehydrated to a milky or cream-colored liquid with small pieces of cereal within 15 to 45 seconds.

i. The color shall be light tan to medium tan, with corn flakes a golden tan.

j. The odor and flavor shall be slightly sweet, slight dairy or creamer, and slight cereal typical of the type of cereal used.

k. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

l. The E. coli count shall be less than 3 per gram (no positive tubes in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to ensure compliance with the requirements of this specification.

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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance of palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Laminated pouch material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraphs	Test procedure
Thickness of films	5.1.1.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.

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Laminated material identification and construction	5.1.1.1.1 and 5.1.1.2.1	Laboratory evaluation.
Total thickness	5.1.1.2.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.

4.5.1.3 Unfilled pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to requirements specified in 5.1.1.1.1. When deemed necessary by the USDA, testing of the unfilled preformed pouches for seal strength and integrity and air tightness of manufacturer's seals shall be as specified in 4.5.1.3.1 and 4.5.1.3.2.

4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1.1 or 5.1.1.2.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.3.2 Unfilled preformed pouch integrity and air tightness of manufacturer's seals testing. Pouch integrity and air tightness of seals shall be determined by placing rigid material similar in size and shape to the bars into the pouch and sealing under a vacuum of not less than 25 inches of mercury. Not less than 96 hours after vacuum sealing, the pouch shall be examined for vacuum loss in accordance with the procedure in 4.5.4. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

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4.5.2.1 In-process examination at the assembly plant. Bulk-packed cereal bars shall be examined before unit packing into individual pouches. The lot size shall be expressed in shipping containers. The sample shall be the number of shipping containers indicated by inspection level S-3. Two bars shall be randomly selected from each of the shipping containers in the sample. Any bar found to have objectionable taste or odor (for example, scorched, rancid, burnt, stale, sour, moldy,) shall be cause for rejection of the lot.

4.5.3 Filled and sealed pouch examination (applicable to unit packed and bulk packed pouches). The filled and sealed pouches shall be examined for defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE I. Filled and sealed pouch defects (applicable to unit packed and bulk packed pouches) 1/

Category		Defect
Major	Minor	
101		Pouch size not as specified 2/
102		Not heat sealed as specified
103		Not clean 3/
104		Tear, hole, or open seal
105		Pouch has foreign odor
106		Evidence of delamination 4/
107		Seal widths not as specified
108		Closure seal not located as specified 5/
109		Presence of foldover wrinkles, occluded matter, and entrapped moisture or grease that reduces the effective closure seal to less than 1/8 inch 5/
110		Required labeling or marking missing, incorrect, illegible, or that smudges

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TABLE I. Filled and sealed pouch defects (applicable to unit packed and bulk packed pouches) 1/ (cont'd)

Category		Defect
Major	Minor	
111		Bars in same pouch more than 1/8 inch out of side by side alignment (style A) 6/
	201	Tear notch missing 6/
	202	Tear notch not located as specified 6/
	203	Depth of tear notch not as specified 6/

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Measurement may be accomplished on unfilled pouches being used for current production lot.

3/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

4/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Pouches exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed by rotating both hands in alternating clockwise-counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the pouch material.

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After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between the thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, it shall be considered evidence of delamination.

5/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/8 inch wide from side seal to side seal that produces a hermetically sealed pouch.

6/ Applicable to unit pouches only.

4.5.4 Pouch vacuum examination. Not less than 96 hours after filling and sealing, the pouches shall be visually examined for conformance to the vacuum requirements in 5.1.1.1.2 and 5.1.1.2.2. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Net-weight examination of product when unit packed in pouches as specified in 5.1.1.1. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 64 grams or greater than 76 grams shall be classified as a minor defect. The lot size shall be expressed in pouches. The results shall be reported to the nearest gram. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.6 Product examination when unit packed as specified in 5.1.1.1. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

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TABLE II. Product defects (unit pack) 1/ 2/

Category		Defect
Major	Minor	
101		Presence of compression streaks or stains
102		Dimensions of bar(s) not as specified
103		Color of bar(s) not as specified
104		Odor and flavor of bar(s) not as specified
105		Bar(s) does not rehydrate as specified in 3.6h
106		Number of bars not as specified (applicable to style A)
	201	Broken or crumbled bars (more than three pieces) 3/

1/ The presence of any foreign material (for example, dirt, insects, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA agent)

3/ A broken or crumbled bar is a bar which either:

a. has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume, or

b. has more than 1/8th of the volume of the bar crushed.

4.5.7 Product testing when unit packs are held over 30 days (see 5.1.1.1). When unit packed bars are held over 30 days before final assembly, the storage container shall be tested for conformance to the oxygen content or vacuum requirements of 5.1.1.1. Testing shall be as specified in 4.5.7.1 or 4.5.7.2. The lot size shall be expressed in number of storage containers. The sample unit shall be one sealed container. The inspection level shall be S-2. Any storage container failing to meet the oxygen content or vacuum requirements shall be cause for rejection of the lot.

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4.5.7.1 Oxygen content testing. The analysis to determine the percentage of oxygen in the nitrogen atmosphere shall be in accordance with any accepted USDA method, using appropriate methods to adapt to the bulk container in which the bars are stored.

4.5.7.2 Vacuum testing. The vacuum level shall be determined by testing in accordance with any accepted USDA method, using appropriate methods to adapt to the bulk container in which the bars are stored.

4.5.8 Examination for weight of product when bulk packed in pouches as specified in 5.1.1.2. The weight of cereal bars shall be determined by individually weighing two bars randomly selected from each sample unit. Any individual bar weight of less than 32 grams or greater than 38 grams for Style A or less than 64 grams or greater than 76 grams for Style B shall be classified as a minor defect. The lot size shall be expressed in bulk packed pouches. The results shall be reported to the nearest gram. The sample unit shall be one bulk packed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units shall be 2.5.

4.5.9 Product examination when bulk packed as specified in 5.1.1.2. The finished product shall be examined for defects listed in table III. The lot size shall be expressed in bulk packed pouches. The sample unit shall be two bars selected at random from the bulk packed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE III. Products defects (bulk pack) 1/ 2/

Category		Defect
Major	Minor	
101		Presence of compression streaks or stains
102		Dimensions of bars not as specified
103		Color of bars not as specified
104		Odor and flavor of bars not as specified
105		Bar(s) not rehydrated as specified in paragraph 3.6h
106		Number of bars not as specified 3/
	201	Broken or crumbled bars (more than three pieces) 4/

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1/ The presence of any foreign material (for example, dirt, insects, insect parts, paper, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), foreign color, shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA agent).

3/ The sample unit shall be the selected bulk packed pouch.

4/ A broken or crumbled bar is a bar which either:

a. has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume, or

b. has more than 1/8th of the volume of the bar crushed.

4.5.10 Pouch closure seal strength testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 inch in width and to a length suitable for proper mounting. Three adjacent specimens shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.2 or 5.1.1.2.2 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-1.

4.5.11 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Official Methods of Analysis of the Association of Official Analtyical Chemists; Chapter: Cereal Foods; Section: Total Solids (Moisture, Indirect Method); Method: Vacuum oven, except that the drying cycle shall be 16 hours at 70°C under a pressure of not more than 100 mm Hg. The test results shall be reported to the nearest 0.1 percent. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed unit packed pouch or two bars selected at random from bulk packed pouches. The inspection level shall be S-2.

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4.5.12 Microbiological testing. Five filled and sealed unit packed pouches or two bars selected from each of five filled and sealed bulk packed pouches shall be selected at random from the lot regardless of lot size. The pouches or two bar units shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Microbiological Methods; Section: Frozen, Chilled, Precooked, or Prepared Foods - Official First Action. The diluent shall be added to each cereal bar and allowed to stand for 15 minutes before blending the sample. Continue with the procedure as directed in the Official First Action. Any result not conforming to the microbiological requirements in 3.6k and 3.6l shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.13 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Number of bars not as specified
Pouches not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The product shall be unit packed in accordance with 5.1.1.1 or bulk packed in accordance with 5.1.1.2.

5.1.1.1 Unit packs. When specified (see 6.1), two cereal bars shall be unit packed in a preformed pouch as specified in 5.1.1.1.1. The unit packed bars shall be stored in a closed container. If bars are held for more than 30 days from date of pack to ration assembly, they shall be protected from oxygen by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen or holding under a vacuum of at least 27 inches of mercury during the period exceeding 30 days. In no case shall be bars be held for more than 90 days from date of pack to ration assembly.

5.1.1.1.1 Preformed pouch for unit pack. The preformed pouches shall be fabricated from inside to outside 0.004 inch thick (± 0.001 inch) polyolefin film laminated to 0.0005 inch thick (± 0.0001 inch) polyester. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having inside

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dimensions of 3 inches wide by 4-7/8 inches long (± 1/16 inch), with the 3 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 1/4 inch (± 1/16 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal strength of not less than 10 pounds per inch and no individual specimen shall have a seal strength of less than 9 pounds per inch when tested as specified in 4.5.1.3.1. A tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch shall be provided with an extended lip extending 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended lip.

5.1.1.1.2 Filling and sealing of unit pack pouches. Two cereal bars of style A or one bar of style B shall be placed into a preformed pouch as specified in 5.1.1.1.1. The filled pouch shall then be hermetically sealed, by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal of not less than 1/8 inch wide. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The closure seal shall not extend below the tear notch on either side of the pouch. The average closure seal strength shall be not less than 10 pounds per inch and no individual test specimen shall be less than 9 pounds per inch when tested as specified in 4.5.10. After sealing, the bars shall remain in an oriented position being no more than 1/8 inch out of side by side alignment (style B). The filled and sealed pouch shall show no loss of vacuum when examined as specified in 4.5.4.

5.1.1.2 Bulk packs. When specified (see 6.1), 30 bars of style A or 15 bars of style B cereal bars shall be bulk packed in a pouch as specified in 5.1.1.2.1. The bulk packed bars shall be held for not more than 90 days from the date of the pack to ration assembly.

5.1.1.2.1 Preformed pouch for bulk pack. The preformed pouch shall be fabricated from a hot fill type, minimum 0.006 inch thick, 3-ply laminate from inside to outside, of polyolefin/aluminum foil/polyester material. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having outside dimensions of 12 inches wide by 15 inches long (± 1/8 inch), with the 12 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal

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strength of not less than 10 pounds per inch and no individual specimen shall have a seal strength of less than 9 pounds per inch when tested as specified in 4.5.1.3.1.

5.1.1.2.2 Filling and sealing of bulk pack pouches. Thirty cereal bars of style A or 15 bars of style B shall be single tier packed into the bulk pack pouch. The filled pouch shall then be hermetically sealed, by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal of not less then 1/8 inch wide. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The average closure seal strength shall be not less than 10 pounds per inch and no individual test specimen shall be less than 9 pounds per inch when tested as specified in 4.5.10. The filled and sealed pouch shall show no loss of vacuum when examined as specified in 4.5.4.

5.2 Packing for shipment to ration assembler. Three hundred of style A or 150 of style B cereal bars, preserved as specified in 5.1, shall be packed flat in layers in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, variety SW, grade 275, class domestic of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick conforming to type I of PPP-C-1797 or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and at the bottom and top of the box to minimize movement of the individual packs.

5.3 Labeling and marking.

5.3.1 Unit pack pouches. Unit pack pouches shall be labeled in a permanent, contrasting color directly on the packaging material in a manner that does not damage the pouches, with a food compatible ink which is free of carcinogenic elements or ingredients. No marking shall be closer than 1/16 inch to any seal. Pressure-sensitive labels may be used instead of marking directly on the packaging material. Labels shall be of a water resistant grade of paper or film, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material shall permit flexibility. Labels shall be suitable for printing or writing on with ink without feathering or spreading. The adhesive shall be of a pressure-sensitive permanent type. The adhesive shall be water insoluble, homogenous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall be of a type that will adhere to the pouch surface under high or low temperatures. Dimensions of the label shall not be larger than the dimensions of the surface of the product where the label is applied on the package. Ink and adhesive used must be suitable for use with food products and shall be free of carcinogenic elements or ingredients. Letter size shall be not less than 1/16 inch high. The following information must be included:

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Item Name
Mixing Instructions
Date of production/lot number (Julian Date) 1/
Name of contractor

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code.

5.3.1.1 Each unit pack shall contain item name:

SHREDDED WHEAT CEREAL BAR	MALTED WHEAT GRANULES CEREAL BAR
CORN FLAKE CEREAL BAR	BRAN FLAKES CEREAL BAR
WHEAT FLAKES CEREAL BAR	OAT CEREAL BISCUIT BAR

5.3.1.2 Mixing instructions are as follows:

Eat as is, or if desired, rehydrate alone or with Dairy bar. To rehydrate, crumble bar, add small amount of water, stir.

5.3.2 Bulk pack pouches. Bulk pack pouches shall be marked as specified in 5.3.1 except that the labels shall include the following information:

Number of bars
Contract number
Name, address, and zip code of contractor

5.3.3 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and style of product required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Whether unit packing or bulk packing is required (see 5.1.1).
f. Provisions for approved preproduction samples (see 6.2).

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6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Technical information.

6.4.1 Malted wheat granules cereal bar, type IV. This product was developed using Grapenuts produced by Post Division of General Foods Corp. Other similar products may give equivalent results.

6.4.2 Oat cereal biscuit bar, type VI. This product was developed using Life cereal produced by Quaker Oats Co. Other similar products may give equivalent results.

6.5 Subject term (key word) listing.

Bar, dehydrated
Operational ration
Rations

Custodian:

Army - GL

Preparing activity:

Army - GL

(Project 8920-A529)

Review activities:

Army - MD, QM
Navy - MC, SA
Air Force - 50
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44389(GL)	Cereal Bars, Ration Lightweight, 30-Day

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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